

APPLICATION OF MACHINE LEARNING CLUSTERING TO THE GROUPING OF STATE INVESTMENT PROJECTS

I. INTRODUCTION

The Bolivian State is responsible for the implementation of 53 industrial projects in different sectors, such as food, industrial inputs, and the pharmaceutical industry. These projects show heterogeneous levels of progress and implementation under varying conditions; however, they require an appropriate classification in order to apply similar measures within the framework of a new industrial policy for the country.

II. GENERAL DESCRIPTION OF THE PROJECTS

The industrial projects being implemented by the Bolivian State in the manufacturing sector, under the responsibility of the Central Government, began during previous administrations within the framework of a policy of strong state intervention. Their purpose was to substitute imports, mainly with regard to consumer goods, and to contribute to the production of industrial inputs, an area in which there is a marked dependence on imported products. This dependence is one of the causes of the disconnection among the different links in the country's production chain. The projects also aim to stimulate regional economies through the integration of agricultural production with industry.

The aforementioned projects present different levels of progress and different characteristics in several areas, including ownership of land and real estate, access to raw materials and markets, financing, debts to third parties, and other factors that make their continuity and viability scenarios more complex.

Furthermore, given the critical situation of the Bolivian economy and, above all, the scarcity of financial resources currently available to the State in both foreign and domestic currency, it is very difficult to continue implementing industrial projects through public financing. It is therefore necessary to seek other sources of funding and to propose alternatives that take into account the prevailing situation and the new governmental vision of incorporating the participation of the national and international private sector in synergy with the State.

III. OBJECTIVES

To apply clustering algorithms to a database of 53 projects with different levels of progress, locations, and characteristics, in order to obtain a compartmentalization based on similar features that allows the application of packages of measures aimed at determining their degree of viability and designing actions related to their diagnosis and classification.

For this specific topic, the K-means algorithm will be applied.

IV. CHARACTERISTICS OF THE PROJECTS

The industrial projects under the responsibility of the Central Government have the following characteristics:

- Different committed investment amounts, ranging from 4 million bolivianos to 3.5 billion bolivianos, in different parts of the country and in different sectors such as food, industrial inputs, and the pharmaceutical industry.
- Different degrees of physical progress, related to payments associated with the progress of the different projects at the national level.
- Debts to third parties, mainly to the Central Bank of Bolivia, with committed interest payments.
- Different degrees of progress regarding ownership rights over the land on which the projects are being implemented, including interinstitutional agreements with municipalities and departmental governments, loan-for-use agreements, municipal laws for land transfer, and property titles registered with the Real Estate Registry.
- Different characteristics in terms of logistical facilities and access to markets and raw materials, with varying degrees of potential.
- Debts to third parties expressed as a percentage of the project amount.
- Strategic positioning of the project, considering its degree of importance in the economy or its impact on the industrial and/or regional sector.

V. K-MEANS ALGORITHM

The K-means clustering algorithm is an unsupervised learning algorithm used for data grouping; it groups unlabeled data points into groups or clusters. The K-means clustering algorithm classifies data points into clusters using a mathematical distance measure, generally Euclidean distance, from the cluster center. The objective is to minimize the sum of distances between data points and their assigned clusters. Data points closest to the centroid are grouped within the same category. A higher value of k, or number of clusters, indicates smaller clusters with greater detail, whereas a lower value of k results in larger clusters with less detail.

The following matrix presents the characteristics of the indicated projects, numbered from 1 to 53.

Table 1. Characteristics of the projects under analysis

Project	Amount in MM of Bs.	Physical progress	FINPRO debt	Raw-material difficulty	Ownership rights	Logistical situation	Market	Debt to third parties as a share of project amount	Strategic situation
Project 1	87,12	72%	87,12	6	6	6	5	0,278	6
Project 2	141,17	79%	141,17	8	5	6	8	0,085	6
Project 3	153,66	79%	153,66	8	5	6	8	0,194	6
Project 4	161,87	64%	0,00	7	5	6	6	0,009	6
Project 5	87,71	33%	87,71	7	8	8	6	0,000	6
Project 6	81,75	83%	81,75	8	5	7	8	0,000	7
Project 7	607,92	0%	0,00	5	3	5	9	0,000	8
Project 8	3.409,90	11%	0,00	8	4	6	8	0,000	10
Project 9	3.348,98	1%	0,00	7	5	8	9	0,000	8
Project 10	45,42	100%	0,00	9	5	7	8	0,000	4
Project 11	24,67	100%	0,00	7	6	9	7	0,000	4
Project 12	45,42	100%	0,00	9	7	8	8	0,000	4
Project 13	5,17	88%	0,00	9	7	6	7	0,427	5
Project 14	182,00	100%	0,00	6	5	10	10	0,015	5
Project 15	162,75	100%	0,00	6	5	9	7	0,050	8
Project 16	424,45	100%	0,00	7	6	8	7	0,118	7
Project 17	106,00	89%	0,00	4	5	5	5	0,000	8
Project 18	87,92	87%	0,00	7	8	5	5	0,000	6
Project 19	83,65	76%	0,00	7	8	7	8	0,000	7
Project 20	65,64	91%	0,00	6	5	8	7	0,035	6
Project 21	260,70	54%	0,00	9	10	6	9	0,077	8

Project 22	77,69	76%	0,00	6	5	7	8	0,005	7
Project 23	62,38	90%	0,00	7	5	8	5	0,003	6
Project 24	45,46	100%	0,00	10	10	9	10		10
Project 25	73,37	100%	0,00	8	7	6	6	0,012	7
Project 26	50,00	71%	0,00	7	5	6	6	0,200	4
Project 27	74,75	20%	0,00	8	5	6	7	0,000	4
Project 28	141,06	25%	0,00	6	6	6	6	0,000	5
Project 29	77,43	20%	0,00	7	5	5	5	0,001	4
Project 30	94,53	11%	0,00	6	5	5	5	0,002	5
Project 31	93,12	23%	0,00	7	4	6	4	0,059	4
Project 32	4,83	0%	0,00	10	10	6	8	0,000	5
Project 33	6,00	59%	0,00	9	7	6	7	0,177	5
Project 34	7,43	20%	0,00	9	7	7	7	0,020	5
Project 35	19,78	51%	0,00	9	7	7	7	0,152	5
Project 36	122,49	60%	0,00	6	6	8	7	0,008	8
Project 37	60,58	65%	0,00	7	5	6	5	0,000	4
Project 38	206,70	50%	0,00	7	4	6	5	0,000	5
Project 39	44,99	64%	0,00	6	5	7	7	0,014	6
Project 40	64,74	65%	0,00	6	5	7	7	0,024	6
Project 41	40,37	53%	0,00	6	5	7	7	0,017	6
Project 42	56,21	44%	0,00	6	5	7	7	0,035	6
Project 43	48,64	39%	0,00	6	5	7	7	0,000	6
Project 44	51,45	48%	0,00	6	5	7	7	0,056	6
Project 45	44,43	47%	0,00	6	5	7	7	0,125	6
Project 46	44,55	41%	0,00	6	5	7	7	0,058	6
Project 47	46,43	16%	0,00	6	5	6	7	0,025	6
Project 48	58,05	56%	0,00	6	5	9	8	0,000	7

Project 49	68,38	48%	0,00	6	5	7	8	0,000	7
Project 50	164,32	34%	0,00	6	5	5	8	0,055	7
Project 51	68,38	45%	0,00	6	5	6	8	0,052	7
Project 52	164,32	25%	0,00	6	5	6	8	0,012	8
Project 53	164,32	2%	0,00	6	5	6	8	0,000	8

Source: Prepared by the author.

The different projects are listed, including as factors explained above some quantitative variables, such as the degree of physical progress, debts to financing entities, and debts to third parties (contractors and supervisors), as well as a rating from 1 to 10 in the remaining categories, with 10 considered the highest score. These assessments were made by personnel familiar with the situation of the projects, based on expertise and expert judgment.

With this information, the analysis categories are normalized proportionally on a scale from 1 to 10 for the corresponding quantitative variables, generating the following table:

Table 2. Characteristics of the projects under analysis

Project	Amount in MM of Bs.	Physical progress	Debt to entities	Raw-material difficulty	Ownership rights	Logistical situation	Market	Debt to third parties as a share of project amount	Strategic situation
Project 1	87,12	7,20	4,33	6	6	6	5	3,48	6
Project 2	141,17	7,90	0,81	8	5	6	8	8,01	6
Project 3	153,66	7,90	0,00	8	5	6	8	5,45	6

Project 4	161,87	6,40	10,00	7	5	6	6	9,78	6
Project 5	87,71	3,30	4,29	7	8	8	6	10,00	6
Project 6	81,75	8,30	4,68	8	5	7	8	10,00	7
Project 7	607,92	0,00	10,00	5	3	5	9	10,00	8
Project 8	3.409,90	1,10	10,00	8	4	6	8	10,00	10
Project 9	3.348,98	0,10	10,00	7	5	8	9	10,00	8
Project 10	45,42	10,00	10,00	9	5	7	8	10,00	4
Project 11	24,67	10,00	10,00	7	6	9	7	10,00	4
Project 12	45,42	10,00	10,00	9	7	8	8	10,00	4
Project 13	5,17	8,80	10,00	9	7	6	7	0,00	5
Project 14	182,00	10,00	10,00	6	5	10	10	9,64	5
Project 15	162,75	10,00	10,00	6	5	9	7	8,82	8
Project 16	424,45	10,00	10,00	7	6	8	7	7,24	7
Project 17	106,00	8,90	10,00	4	5	5	5	10,00	8
Project 18	87,92	8,70	10,00	7	8	5	5	10,00	6
Project 19	83,65	7,60	10,00	7	8	7	8	10,00	7
Project 20	65,64	9,10	10,00	6	5	8	7	9,17	6
Project 21	260,70	5,40	10,00	9	10	6	9	8,20	8
Project 22	77,69	7,60	10,00	6	5	7	8	9,88	7
Project 23	62,38	9,00	10,00	7	5	8	5	9,92	6
Project 24	45,46	10,00	10,00	10	10	9	10	10,00	10
Project 25	73,37	10,00	10,00	8	7	6	6	9,71	7
Project 26	50,00	7,10	10,00	7	5	6	6	5,32	4
Project 27	74,75	2,00	10,00	8	5	6	7	10,00	4
Project 28	141,06	2,50	10,00	6	6	6	6	10,00	5
Project 29	77,43	2,00	10,00	7	5	5	5	9,98	4
Project 30	94,53	1,10	10,00	6	5	5	5	9,96	5

Project 31	93,12	2,30	10,00	7	4	6	4	8,61	4
Project 32	4,83	0,00	10,00	10	10	6	8	10,00	5
Project 33	6,00	5,90	10,00	9	7	6	7	5,86	5
Project 34	7,43	2,00	10,00	9	7	7	7	9,53	5
Project 35	19,78	5,10	10,00	9	7	7	7	6,43	5
Project 36	122,49	6,00	10,00	6	6	8	7	9,81	8
Project 37	60,58	6,50	10,00	7	5	6	5	10,00	4
Project 38	206,70	5,00	10,00	7	4	6	5	10,00	5
Project 39	44,99	6,40	10,00	6	5	7	7	9,68	6
Project 40	64,74	6,50	10,00	6	5	7	7	9,44	6
Project 41	40,37	5,30	10,00	6	5	7	7	9,60	6
Project 42	56,21	4,40	10,00	6	5	7	7	9,18	6
Project 43	48,64	3,90	10,00	6	5	7	7	10,00	6
Project 44	51,45	4,80	10,00	6	5	7	7	8,68	6
Project 45	44,43	4,70	10,00	6	5	7	7	7,07	6
Project 46	44,55	4,10	10,00	6	5	7	7	8,64	6
Project 47	46,43	1,60	10,00	6	5	6	7	9,41	6
Project 48	58,05	5,60	10,00	6	5	9	8	10,00	7
Project 49	68,38	4,80	10,00	6	5	7	8	10,00	7
Project 50	164,32	3,40	10,00	6	5	5	8	8,71	7
Project 51	68,38	4,50	10,00	6	5	6	8	8,78	7
Project 52	164,32	2,50	10,00	6	5	6	8	9,71	8
Project 53	164,32	0,20	10,00	6	5	6	8	10,00	8

Source: Prepared by the author.

The following script in R is used for the classification:

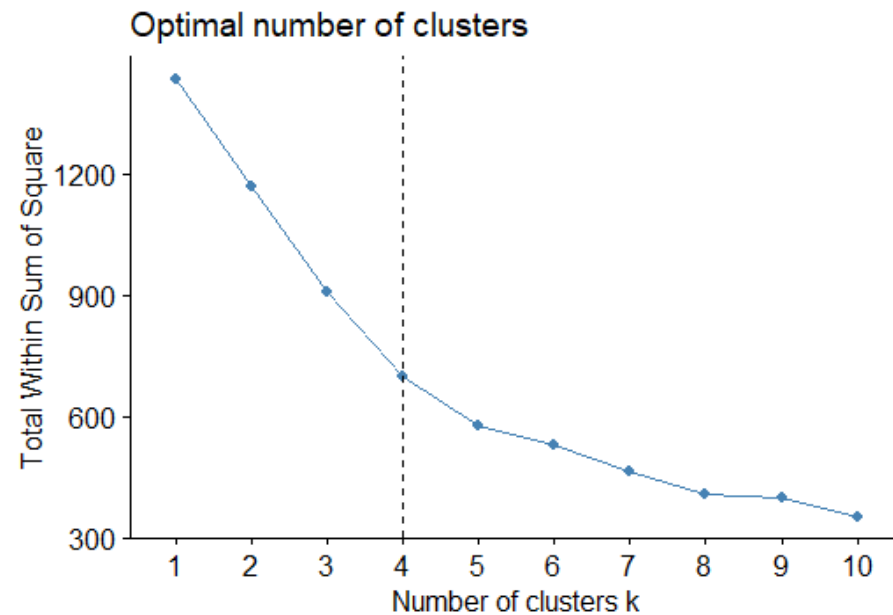
```
install.packages("readxl")
library(readxl)
install.packages("ggplot2")
library(ggplot2)
install.packages("factoextra")
library("factoextra")
datos <- read_excel("C:\\Users\\SIWI II\\Downloads\\para conglomerados.xlsx", sheet = "Pesos2 conc")
View(datos)
datosConglom <- datos[, 3:10]
head(datosConglom)
View(datosConglom)
fviz_nbclust(datosConglom, kmeans, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2)
set.seed(123)
k <- 4
km.res <- kmeans(datosConglom, centers = k, nstart = 25)
print(km.res)
```



```
df.clasif <- cbind(datosConglom, cluster = km.res$cluster)
head(datosConglom)
View(df.clasif)
```

Figure 1. Decision on the number of clusters

As can be observed in the elbow plot used to determine the optimal number of clusters, it is advisable to consider 4 and 5 as possible values.



Source: Prepared by the author based on the K-means algorithm.

This algorithm yields an efficiency indicator (between-cluster distance / total distance = 52.6%), which is a good result.

The following classification is generated:

Table 3. Classification of projects with k = 4

Project	Amount in MM of Bs.	Physical progress	Debt to entities	Raw-material difficulty	Ownership rights	Logistical situation	Market	Debt to third parties as a share of project amount	Strategic situation	CLUSTER
Project 13	5,17	8,80	10,00	9,00	7,00	6,00	7,00	0,00	5,00	1,00
Project 21	260,70	5,40	10,00	9,00	10,00	6,00	9,00	8,20	8,00	1,00
Project 26	50,00	7,10	10,00	7,00	5,00	6,00	6,00	5,32	4,00	1,00
Project 33	6,00	5,90	10,00	9,00	7,00	6,00	7,00	5,86	5,00	1,00
Project 35	19,78	5,10	10,00	9,00	7,00	7,00	7,00	6,43	5,00	1,00
Project 4	161,87	6,40	10,00	7,00	5,00	6,00	6,00	9,78	6,00	2,00
Project 10	45,42	10,00	10,00	9,00	5,00	7,00	8,00	10,00	4,00	2,00
Project 11	24,67	10,00	10,00	7,00	6,00	9,00	7,00	10,00	4,00	2,00
Project 12	45,42	10,00	10,00	9,00	7,00	8,00	8,00	10,00	4,00	2,00
Project 14	182,00	10,00	10,00	6,00	5,00	10,00	10,00	9,64	5,00	2,00
Project 15	162,75	10,00	10,00	6,00	5,00	9,00	7,00	8,82	8,00	2,00
Project 16	424,45	10,00	10,00	7,00	6,00	8,00	7,00	7,24	7,00	2,00
Project 17	106,00	8,90	10,00	4,00	5,00	5,00	5,00	10,00	8,00	2,00
Project 18	87,92	8,70	10,00	7,00	8,00	5,00	5,00	10,00	6,00	2,00
Project 19	83,65	7,60	10,00	7,00	8,00	7,00	8,00	10,00	7,00	2,00
Project 20	65,64	9,10	10,00	6,00	5,00	8,00	7,00	9,17	6,00	2,00
Project 22	77,69	7,60	10,00	6,00	5,00	7,00	8,00	9,88	7,00	2,00

Project 23	62,38	9,00	10,00	7,00	5,00	8,00	5,00	9,92	6,00	2,00
Project 24	45,46	10,00	10,00	10,00	10,00	9,00	10,00	10,00	10,00	2,00
Project 25	73,37	10,00	10,00	8,00	7,00	6,00	6,00	9,71	7,00	2,00
Project 36	122,49	6,00	10,00	6,00	6,00	8,00	7,00	9,81	8,00	2,00
Project 37	60,58	6,50	10,00	7,00	5,00	6,00	5,00	10,00	4,00	2,00
Project 39	44,99	6,40	10,00	6,00	5,00	7,00	7,00	9,68	6,00	2,00
Project 40	64,74	6,50	10,00	6,00	5,00	7,00	7,00	9,44	6,00	2,00
Project 48	58,05	5,60	10,00	6,00	5,00	9,00	8,00	10,00	7,00	2,00
Project 1	87,12	7,20	4,33	6,00	6,00	6,00	5,00	3,48	6,00	3,00
Project 2	141,17	7,90	0,81	8,00	5,00	6,00	8,00	8,01	6,00	3,00
Project 3	153,66	7,90	0,00	8,00	5,00	6,00	8,00	5,45	6,00	3,00
Project 5	87,71	3,30	4,29	7,00	8,00	8,00	6,00	10,00	6,00	3,00
Project 6	81,75	8,30	4,68	8,00	5,00	7,00	8,00	10,00	7,00	3,00
Project 7	607,92	0,00	10,00	5,00	3,00	5,00	9,00	10,00	8,00	4,00
Project 8	3.409,90	1,10	10,00	8,00	4,00	6,00	8,00	10,00	10,00	4,00
Project 9	3.348,98	0,10	10,00	7,00	5,00	8,00	9,00	10,00	8,00	4,00
Project 27	74,75	2,00	10,00	8,00	5,00	6,00	7,00	10,00	4,00	4,00
Project 28	141,06	2,50	10,00	6,00	6,00	6,00	6,00	10,00	5,00	4,00
Project 29	77,43	2,00	10,00	7,00	5,00	5,00	5,00	9,98	4,00	4,00
Project 30	94,53	1,10	10,00	6,00	5,00	5,00	5,00	9,96	5,00	4,00
Project 31	93,12	2,30	10,00	7,00	4,00	6,00	4,00	8,61	4,00	4,00
Project 32	4,83	0,00	10,00	10,00	10,00	6,00	8,00	10,00	5,00	4,00
Project 34	7,43	2,00	10,00	9,00	7,00	7,00	7,00	9,53	5,00	4,00
Project 38	206,70	5,00	10,00	7,00	4,00	6,00	5,00	10,00	5,00	4,00
Project 41	40,37	5,30	10,00	6,00	5,00	7,00	7,00	9,60	6,00	4,00
Project 42	56,21	4,40	10,00	6,00	5,00	7,00	7,00	9,18	6,00	4,00

Project 43	48,64	3,90	10,00	6,00	5,00	7,00	7,00	10,00	6,00	4,00
Project 44	51,45	4,80	10,00	6,00	5,00	7,00	7,00	8,68	6,00	4,00
Project 45	44,43	4,70	10,00	6,00	5,00	7,00	7,00	7,07	6,00	4,00
Project 46	44,55	4,10	10,00	6,00	5,00	7,00	7,00	8,64	6,00	4,00
Project 47	46,43	1,60	10,00	6,00	5,00	6,00	7,00	9,41	6,00	4,00
Project 49	68,38	4,80	10,00	6,00	5,00	7,00	8,00	10,00	7,00	4,00
Project 50	164,32	3,40	10,00	6,00	5,00	5,00	8,00	8,71	7,00	4,00
Project 51	68,38	4,50	10,00	6,00	5,00	6,00	8,00	8,78	7,00	4,00
Project 52	164,32	2,50	10,00	6,00	5,00	6,00	8,00	9,71	8,00	4,00
Project 53	164,32	0,20	10,00	6,00	5,00	6,00	8,00	10,00	8,00	4,00

Source: Prepared by the author based on the K-means algorithm.

The analysis is then conducted for 5 clusters, yielding an efficiency indicator (between-cluster distance / total distance = 59.8%), which is a good result.

The following classification is generated:

Table 4. Classification of projects with k = 5

Project	Amount in MM of Bs.	Physical progress	Debt to entities	Raw-material difficulty	Ownership rights	Logistical situation	Market	Debt to third parties as a share of project amount	Strategic situation	CLUSTER
Project 7	607,92	0,00	10,00	5,00	3,00	5,00	9,00	10,00	8,00	1,00
Project 8	3.409,90	1,10	10,00	8,00	4,00	6,00	8,00	10,00	10,00	1,00
Project 9	3.348,98	0,10	10,00	7,00	5,00	8,00	9,00	10,00	8,00	1,00
Project 27	74,75	2,00	10,00	8,00	5,00	6,00	7,00	10,00	4,00	1,00

Project 28	141,06	2,50	10,00	6,00	6,00	6,00	6,00	10,00	5,00	1,00
Project 29	77,43	2,00	10,00	7,00	5,00	5,00	5,00	9,98	4,00	1,00
Project 30	94,53	1,10	10,00	6,00	5,00	5,00	5,00	9,96	5,00	1,00
Project 31	93,12	2,30	10,00	7,00	4,00	6,00	4,00	8,61	4,00	1,00
Project 32	4,83	0,00	10,00	10,00	10,00	6,00	8,00	10,00	5,00	1,00
Project 33	6,00	5,90	10,00	9,00	7,00	6,00	7,00	5,86	5,00	1,00
Project 35	19,78	5,10	10,00	9,00	7,00	7,00	7,00	6,43	5,00	1,00
Project 47	46,43	1,60	10,00	6,00	5,00	6,00	7,00	9,41	6,00	1,00
Project 50	164,32	3,40	10,00	6,00	5,00	5,00	8,00	8,71	7,00	1,00
Project 52	164,32	2,50	10,00	6,00	5,00	6,00	8,00	9,71	8,00	1,00
Project 53	164,32	0,20	10,00	6,00	5,00	6,00	8,00	10,00	8,00	1,00
Project 13	5,17	8,80	10,00	9,00	7,00	6,00	7,00	0,00	5,00	2,00
Project 21	260,70	5,40	10,00	9,00	10,00	6,00	9,00	8,20	8,00	2,00
Project 26	50,00	7,10	10,00	7,00	5,00	6,00	6,00	5,32	4,00	2,00
Project 34	7,43	2,00	10,00	9,00	7,00	7,00	7,00	9,53	5,00	2,00
Project 36	122,49	6,00	10,00	6,00	6,00	8,00	7,00	9,81	8,00	2,00
Project 4	161,87	6,40	10,00	7,00	5,00	6,00	6,00	9,78	6,00	3,00
Project 17	106,00	8,90	10,00	4,00	5,00	5,00	5,00	10,00	8,00	3,00
Project 22	77,69	7,60	10,00	6,00	5,00	7,00	8,00	9,88	7,00	3,00
Project 37	60,58	6,50	10,00	7,00	5,00	6,00	5,00	10,00	4,00	3,00
Project 38	206,70	5,00	10,00	7,00	4,00	6,00	5,00	10,00	5,00	3,00
Project 39	44,99	6,40	10,00	6,00	5,00	7,00	7,00	9,68	6,00	3,00
Project 40	64,74	6,50	10,00	6,00	5,00	7,00	7,00	9,44	6,00	3,00
Project 41	40,37	5,30	10,00	6,00	5,00	7,00	7,00	9,60	6,00	3,00
Project 42	56,21	4,40	10,00	6,00	5,00	7,00	7,00	9,18	6,00	3,00

Project 43	48,64	3,90	10,00	6,00	5,00	7,00	7,00	10,00	6,00	3,00
Project 44	51,45	4,80	10,00	6,00	5,00	7,00	7,00	8,68	6,00	3,00
Project 45	44,43	4,70	10,00	6,00	5,00	7,00	7,00	7,07	6,00	3,00
Project 46	44,55	4,10	10,00	6,00	5,00	7,00	7,00	8,64	6,00	3,00
Project 48	58,05	5,60	10,00	6,00	5,00	9,00	8,00	10,00	7,00	3,00
Project 49	68,38	4,80	10,00	6,00	5,00	7,00	8,00	10,00	7,00	3,00
Project 51	68,38	4,50	10,00	6,00	5,00	6,00	8,00	8,78	7,00	3,00
Project 10	45,42	10,00	10,00	9,00	5,00	7,00	8,00	10,00	4,00	4,00
Project 11	24,67	10,00	10,00	7,00	6,00	9,00	7,00	10,00	4,00	4,00
Project 12	45,42	10,00	10,00	9,00	7,00	8,00	8,00	10,00	4,00	4,00
Project 14	182,00	10,00	10,00	6,00	5,00	10,00	10,00	9,64	5,00	4,00
Project 15	162,75	10,00	10,00	6,00	5,00	9,00	7,00	8,82	8,00	4,00
Project 16	424,45	10,00	10,00	7,00	6,00	8,00	7,00	7,24	7,00	4,00
Project 18	87,92	8,70	10,00	7,00	8,00	5,00	5,00	10,00	6,00	4,00
Project 19	83,65	7,60	10,00	7,00	8,00	7,00	8,00	10,00	7,00	4,00
Project 20	65,64	9,10	10,00	6,00	5,00	8,00	7,00	9,17	6,00	4,00
Project 23	62,38	9,00	10,00	7,00	5,00	8,00	5,00	9,92	6,00	4,00
Project 24	45,46	10,00	10,00	10,00	10,00	9,00	10,00	10,00	10,00	4,00
Project 25	73,37	10,00	10,00	8,00	7,00	6,00	6,00	9,71	7,00	4,00
Project 1	87,12	7,20	4,33	6,00	6,00	6,00	5,00	3,48	6,00	5,00
Project 2	141,17	7,90	0,81	8,00	5,00	6,00	8,00	8,01	6,00	5,00
Project 3	153,66	7,90	0,00	8,00	5,00	6,00	8,00	5,45	6,00	5,00
Project 5	87,71	3,30	4,29	7,00	8,00	8,00	6,00	10,00	6,00	5,00
Project 6	81,75	8,30	4,68	8,00	5,00	7,00	8,00	10,00	7,00	5,00

Source: Prepared by the author.

VI. ANALYSIS OF RESULTS

Once the classification into different compartments has been performed, the division of the projects into 5 clusters is considered more appropriate. In addition, according to the expertise of those familiar with the projects, these clusters group projects with similar characteristics, as described below.

Cluster 1

These are the projects with the greatest implementation difficulties, the lowest levels of physical progress, and problems related to logistics, ownership rights, and raw-material supply. They represent those with the lowest viability and possibility of completion, given the current restrictions on public resources. Likewise, they show low potential for partnerships with private-sector actors.

This cluster includes 15 projects and represents an investment amount of 8,417 million bolivianos.

Cluster 2

These are projects that generally have a low to medium level of progress, with problems related to logistics, debts to third parties, and ownership rights. They have a medium strategic positioning in terms of their impact on national industry, with moderate clarity regarding their continuity viability and potential for a strategic partnership with the private sector.

This cluster includes 6 projects and represents an investment amount of 446 million bolivianos.

Cluster 3

These are projects that generally have a medium-high level of progress; however, it must be verified whether such progress is homogeneous in both infrastructure and equipment. They present problems related to ownership rights. Their situation regarding raw materials and markets is similar to that of the previous cluster, and their strategic importance for the industry is low. They show moderate potential for continuity and for partnerships with the private sector to support their viability.

This cluster includes 16 projects and represents an investment amount of 1,203 million bolivianos.

Cluster 4

These are projects that are generally completed or have a high degree of progress. They present problems related to ownership rights; their strategic importance for the industry is medium-high, and they show high

potential for continuity and for partnerships with the private sector to support their viability.

This cluster includes 12 projects and represents an investment amount of 1,303 million bolivianos.

Cluster 5

These are projects with a high degree of progress. They present problems related to ownership rights; their strategic importance for the industry is medium-high, and they show high potential for continuity and for partnerships with the private sector to support their viability. This group is similar to Cluster 4, although some of these projects have debts to third parties and all of them have debts to financial entities.

This cluster includes 5 projects and represents an investment amount of 551 million bolivianos.

VII. CONCLUSIONS AND RECOMMENDATIONS

The K-means algorithm provides results that are aligned with the criteria currently used by the entity responsible for monitoring these projects, with slight differences. A classification into 5 categories is considered appropriate, as it allows recommendations to be made for each category with the aim of establishing the projects' viability within the framework of the prevailing restrictions.

In this regard, for projects that have been completed and/or that show a high degree of physical progress, and which are practically all located in Cluster 4, partnerships with the private sector should be established to ensure that they operate at maximum capacity. One possible mechanism for this purpose is the sale of processing services or the concession of facilities to private companies, with the approval of the municipalities that own the land.

For projects that are in conditions similar to those described in the previous paragraph, with a high degree of progress but with debts to financial entities, and which correspond to Cluster 5, partnerships with the private sector should likewise be established to ensure that they operate at maximum capacity. One possible mechanism for this purpose is the sale of processing services or the concession of facilities to private companies, with the approval of the municipalities that own the land, as well as the achievement of deferments and/or extensions for the payment of outstanding loans.

For projects with a level of progress above 60%, provided that such progress includes both infrastructure and machinery, it is considered appropriate to complete the investments. To this end, partnerships with

the private sector should be promoted in order to achieve a level of operability in the markets and to obtain operating capital for that purpose, with the prior approval of the municipalities, which in most cases own the land. These projects are mainly located in Cluster 3.

Finally, for the projects located in Clusters 1 and 2—except for those identified as having progress above 60% in both machinery and equipment, which could be treated in the same way as the projects in the previous clusters—it is advisable to consider discontinuation and reformulation.

References

- Information from the Plurinational State of Bolivia.
- <https://www.ibm.com/mx-es/think/topics/k-means-clustering>